Explainable Self-Correcting Multimodal GraphRAG with Adaptive Query Routing for Intelligent Document Question Answering

Abstract

Retrieval-Augmented Generation (RAG) has emerged as an effective approach for question answering over large document repositories by combining information retrieval with Large Language Models (LLMs). However, conventional RAG systems relying solely on vector databases often struggle with multi-hop reasoning, multimodal information retrieval, answer reliability, and transparency. To address these limitations, this paper proposes an **Explainable Self-Correcting Multimodal GraphRAG Framework** for intelligent document question answering. The system constructs a knowledge graph from document structure and content, enabling relationship-aware retrieval across documents, chapters, sections, and semantic entities. It extends traditional GraphRAG by incorporating multimodal knowledge extraction from text, tables, diagrams, and images, transformed into interconnected graph representations for enhanced retrieval and reasoning.

To improve answer reliability, a Critic Agent validates generated responses against retrieved evidence and triggers iterative retrieval and regeneration when inconsistencies or insufficient evidence are detected. A Cascaded Dynamic Retrieval mechanism routes simple factual queries to lightweight vector-based retrieval while directing complex multi-hop queries to the GraphRAG pipeline, reducing latency and computational overhead. For user trust and transparency, the framework provides explainable reasoning traces including source documents, page references, retrieved chunks, graph traversal paths, and supporting evidence. An interactive Knowledge Graph Visualization Dashboard enables users to explore document relationships and observe retrieval pathways in real time.

The framework is particularly suitable for technical domains such as factory planning, manufacturing, maintenance support, and enterprise knowledge management, where information is distributed across large collections of documents, tables, and engineering diagrams. By combining multimodal retrieval, self-correcting answer generation, adaptive query routing, explainability, and graph visualization, the proposed system aims to improve answer accuracy, reliability, scalability, and user trust compared to traditional RAG and GraphRAG approaches.